

Program: Diploma in Fire Technology and Safety	
Course Code: 6462A	Course Title: Safety Management
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the fundamentals of safety engineering and the goals of safety management.
- To familiarize students with accident prevention methods and their role in industrial safety.
- To equip students with knowledge of personal protective equipment (PPE), performance monitoring, and safety inspections.
- To introduce safety management systems, accident investigation processes, and their applications in preventing industrial accidents.

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the principles of safety engineering, accident causation theories, and safety organization roles.	12	Understanding
CO2	Apply accident prevention methods and safety education strategies in an industrial environment.	15	Applying
CO3	Identify the types of PPE, monitor safety performance, and conduct safety inspections.	17	Analyzing
CO4	Investigate accidents, use analytical techniques for accident analysis, and apply safety management systems.	14	Evaluating
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3				
CO2	3	3	2				
CO3	3	3	2				
CO4	3	2	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand the principles of safety engineering, accident causation theories, and safety organization roles.		
M1.01	Definition of safety and its goals in engineering contexts.	3	Understanding
M1.02	Need for safety in industry and its impact on productivity.	3	Understanding
M1.03	History of safety movement and theories of accident causation.	3	Understanding
M1.04	Safety organization: Types, objectives, functions, and roles of management, supervisors, and workers.	3	Applying
Contents: Safety Engineering: Understanding the definition, goals, and importance of safety. Accident Causation: Examining theories that explain the causes of accidents. Safety Organizations: Structure and function of safety committees and roles of key stakeholders like management, workers, and unions. Safety Policies: Developing and implementing safety policies within organizations.			
CO2	Apply accident prevention methods and safety education strategies in an industrial environment.		
M2.01	Methods of accident prevention: Engineering, education, and enforcement.	3	Applying
M2.02	Safety education and training: Importance, methods, and effectiveness.	3	Applying

M2.03	Behavior-based safety: Concept and application in safety management.	3	Understanding
M2.04	Communication in safety: Purpose, barriers, and solutions.	3	Applying
M2.05	Housekeeping in safety: Responsibilities and advantages. 5S of housekeeping.	3	Understanding
	SeriesTestI	1	
Contents: Accident Prevention: Methods for preventing accidents through engineering, education, and enforcement. Safety Education: Training employees in safety protocols and evaluating the effectiveness of these training programs. Behavior-Based Safety: Understanding and applying behavior-based safety programs. Communication in Safety: The importance of clear communication and overcoming barriers in industrial environments. Housekeeping: Best practices for maintaining a safe working environment.			
CO3	Identify the types of PPE, monitor safety performance, and conduct safety inspections.		
M3.01	Types of personal protective equipment (PPE) and standards.	3	Understanding
M3.02	Monitoring safety performance: Frequency, severity, and incidence rates.	3	Analyzing
M3.03	Cost of accidents: Computation and utility of cost data.	5	Analyzing
M3.04	Plant safety inspections: Types, procedures, and safety sampling techniques.	3	Applying
M3.05	Job safety analysis (JSA) and safety audits.	3	Analyzing
Contents: PPE in Industrial Settings: Understanding the types and standards for personal protective equipment. Safety Performance Metrics: Tracking accident frequency, severity, and other relevant safety metrics. Cost of Accidents: Analyzing and computing the costs associated with industrial accidents. Safety Inspections and Audits: Conducting safety inspections, sampling, and analysis techniques.			

CO4	Learn about the design of harbors and docks, and understand the safety features of water transportation systems.		
M4.01	Importance of accident investigation: Why, when, where, who, and how?	3	Understanding
M4.02	Accident investigation process: Tools, data collection, and handling witnesses.	4	Applying
M4.03	Accident analysis: Analytical techniques like MORT, change analysis, and system safety.	4	Analyzing
M4.04	Case study of an accident investigation.	3	Evaluating
	SeriesTest II	1	
Contents: Accident Investigation: Techniques for investigating industrial accidents and understanding the root causes. Analytical Techniques: Use of methods like MORT (Management Oversight and Risk Tree), change analysis, and multi-event sequencing to analyze accidents. Case Study: Practical application of accident investigation techniques through real-world case studies.			

Text/Reference:

T/R	BookTitle/Author
R1	Krishnan, N.V. Safety Management in Industry, Jaico Publishing House, New Delhi, 1997.
R2	Grimaldi, John V., and Simonds, Rollin H. Safety Management, All India Traveller Book Seller, Delhi, 1989.
R3	Blake, Ronald P. Industrial Safety, Prentice Hall, New Delhi, 1973.
R4	Waring, Alan. Safety Management Systems, Chapman & Hall, England, 1996.
R5	National Safety Council Accident Prevention Manual for Industrial Operations, Chicago, 1982.